Information Retrieval: Assignment 1

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1 Introduction

In this assignment, I implemented CV classification application for recruitment result based on embedding model(sentence-transformers/all-MiniLM-L6-v2). The data is based on recruitment data for fictional IT company. This application consists of 2 main components: classifier for topic which is relevant to given sentence or keywords, and semantic CV search given sentence, which gives top 3 the most relative CV. Github link[1], HuggingFace dataset[2], working demo[3] are mentioned in footnote in this page.

2 Dataset

The dataset is generated by Google Antigravity, given prompt for generating the data as context. The dataset contains 10,000 generated recruitment records for a fictional IT company from 2020 to 2026. Among 10,000 data, 8,000 are used as training data, and rest of 2,000 are used as test data.

3 Model

3.1 Embedding model

I chose sentence-transformers/all-MiniLM-L6-v2 as a embedding model, considering balance between efficiency and performance. Also, the application has functionality not only as classifier for recruitment result, but also semantic search system for finding the most similar historical candidates through cosine similarity therefore, I chose the embedding model so that the embedded information could be utilized for the semantic search system.

3.2 Classifier

In order to train several classifiers so that I could choose the best classifier for launching the application, I tried to train Support Vector Classification[4], Random Forest Classifier[5],

[1]https://github.com/katsufu/UUIR-assignment1.git
[2]https://huggingface.co/datasets/katsufu/recruitment_data
[3]https://huggingface.co/spaces/uu-ir-kafu0614/UUIR_Classification_Demo
[4]https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html
[5]https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

and neural network architecture with PyTorch, and evaluate them with F1-score[6]. I chose PyTorch Neural Network, given below result for both Accuracy and F1-score.

Table 1: Model Performance Comparison

Model	**Accuracy**	**F1-Score**
Random Forest	76.7%	76.6%
Support Vector Machine	77.4%	78.1%
PyTorch Neural Network	80.8%	78.5%

4 Application

As above mentioned, the application consists of 2 main components: classifier and semantic search. Given input CV or example one on Candidate CV field, the model evaluate if the CV would be accepted or not as recruitment result. In addition to this, the application suggests top 3 the most similar historical candidates, including position name, recruitment result, the reason for the result, experience years, and excerpt from CV.

5 Reflection

In light of current technological trends, utilizing artificial intelligence for this assignment is a highly practical approach. Specifically, while providing well-contextualized prompts is relatively straightforward for both technical and non-technical individuals, the subsequent need to revise implementation plans and modify source code across multiple file formats remains a significant challenge for non-engineers. I consider this demonstrates why it remains still difficult for AI to fully replace engineering and IT-related professions.

[6] https://scikit-learn.org/stable/modules/generated/sklearn.metrics.f1_score.html